

Technologies that will shape the market

The electronics industry is going through an exciting phase due to revolutionary changes in technology, launch of innovative products and the challenge of global competition. This has made it necessary for electronics product and component manufacturers to focus on continuous improvements in order to stay ahead of the pack. Survey participants have shared insights about emerging technology trends that will shape the market in India, and create more efficient, user-friendly products by using better production techniques. Here is a collation of their views and our analysis.

Evolution of components and products. Miniaturisation. This refers to the creation of smaller devices or components for mechanical, optical and electronic products. Convergence is helping manufacturers integrate multiple devices into one. Demand from consumers to reduce the size of the products so that these are easy to manage has also led to these products becoming smaller in size.

The greater density of components in these products is usually made possible through VLSI designs. This also enables a lower cost of production, which translates to lower overall product pricing. Miniaturisation is on the rise and will impact the traditional components market as most of these components will get replaced by chip components and integrated circuits (ICs).

Artificial intelligence. Consumers are becoming increasingly technology-conscious and are demanding products with built-in intelligence. This is resulting in electronics and consumer durable products being manufactured with intelligent functions and logic. For example, there are now washing machines that can sense the load and decide the appropriate washing cycles.

Intelligence has moved beyond consumer products, and is also avail-

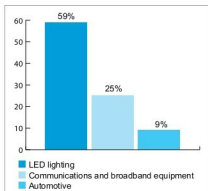


Fig. 2: Top three demand-generating sectors

able in several medical and industrial electronic products with CNC-controlled functions.

High-frequency applications. Electronic ICs and modules for high-frequency applications will be in demand considering the increasing amounts of data that need to be transmitted within a very short period of time, whether in applications for communications, sensors or astronautics.

Advances in IC technology are driving packaging and interconnect designers to accommodate more input/output connections and larger-sized dies, which dissipate more power and operate at higher speeds. This, in turn, will generate demand for components and products suitable for high-frequency applications.

Leading changes in LEDs. There will be changes in onboard technology due to the use of IC based drivers to support touch based technology, which is the need of the hour for smart lighting systems. This, in turn, will reduce the number of components used in traditional drivers. Use of fewer components can reduce costs, while enhancing the efficiency of the final product.

There will be higher usage of chip-scale packaging or flip-chip packaging technology to enhance lumen output while increasing the reliability of the final product. Use of chip-scale packaging eliminates the traditional sub-mount, enabling manufacturers to directly attach the

LED die to the PCB, allowing for overall system cost reductions.

Introduction of driverless low-voltage direct-current-operated products will enable energy savings by reducing AC-DC current conversion losses. This will also make the products compatible with solar photovoltaic systems, and help these run as LED-solar hybrid systems, which will be very useful in India.

Shift in manufacturing techniques from through-hole to surface-mount technology will enhance efficiency while reducing operational costs. This, in turn, will help achieve a breakeven point quickly, in spite of relatively higher capital expenditure.

The electronics behind the IoT. Each IoT device will require, at the very minimum, a microcontroller to add intelligence to it, one or more sensor(s) to allow for data collection, one or more chip(s) for connectivity and data transmission, and a memory component. Connected devices that transmit information across relevant networks will rely on innovations from semiconductor players—highly-integrated microchip designs, for instance, and very low-power functions in certain applications.

A new class of system(s)-on-chip (SoC) based devices with the provision for optimal power and connectivity features as well as with sensor integration will be in demand to support the wide adoption of the IoT. The first generation of such chips is already on the way, although it will probably be a few generations before chips can deliver all the required functionality.

A new array of sensors based on micro-electromechanical systems technology are rapidly being developed to enable IoT applications beyond motion and image sensing to include those that measure humidity, altitude, food calorie composition and human health indicators. This is the result of an increasing variety of consumer IoT applications such as wearables and clustered systems.